

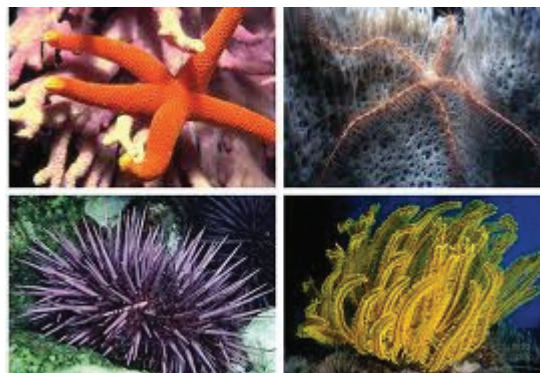
3.9 PHYLUM-ECHINODERMATA

- Deuterostomes with radial bodies

3.9.1 General Characters

The **echinoderms** (Gr. *echinos* : spiny; *dermos* : skin) are non-chordate deuterostomes and enterocoelomates. All are marine. The adult echinoderms are **radially symmetrical** (**pentamerous radial symmetry**), but the larvae are **bilaterally symmetrical**. Body wall is thick and body surface spiny; echinoderms possess a mesodermal endoskeleton of calcareous ossicles, which support the spines, hence the name 'Echinodermata' (*spiny bodied animals*). In some, the spines are modified into '**pedicellariae**'. The most distinctive feature of echinoderms is the presence of **water vascular system** or **ambulacral system** derived from the coelom which helps in locomotion, capture and transport of food and respiration. Water enters the ambulacral system through the *madreporite*. Circulatory system is of **open type**, without heart and blood vessels. Respiratory organs are '**papulae**' or '**dermal branchiae**' (thin outgrowths of the body wall between the spines), **cloacal respiratory trees** (*sea cucumbers*), etc. Special excretory organs are absent (excretion is by simple diffusion). Nervous system is poorly developed and brain is absent; sense organs are poorly developed. Sexes are separate (dioecious) but do not exhibit sexual dimorphism. Reproduction is sexual. Fertilisation is usually external. Development is indirect with a free-swimming **bilaterally symmetrical larva**. Power of regeneration is remarkable. Some exhibit **autotomy** (*auto* : self; *tome* : cut; self mutilation or self amputation) when captured by a predator, and the lost body parts are regenerated. A hypothetical larva, **dipleurula** is considered the ancestor of all echinoderms.

Examples: *Asterias*, *Echinus*, *Antedon*, *Cucumaria* and *Ophiura*.



3.9.2 Classification of the phylum Echinodermata

Phylum Echinodermata is classified into **two sub-phyla** namely **1. Pelmatozoa** and **2. Eleutherozoa**.

Sub-phylum-1: Pelmatozoa

They are sessile, attached to the substratum by long stalk (*sea lilies*) and some are free swimming (*feather stars*). Mouth and anus are present on the upper surface (oral surface). Madreporite is absent. Ambulacral grooves are open.

The sub-phylum **Pelmatozoa** includes **one class only 1. Crinoidea**

Class-1: Crinoidea (Gr. *crinon* : lily; *eidos* : form)

This class includes the oldest and most *primitive* forms. Some are stalk-less and free moving and others are stalked. Viscera is enclosed in a calcareous *test* called **theca**. Arms are **biramous** and bear **pinnules** and help in food capture. Mouth and anus are

present on the upper surface (oral surface). Ambulacral grooves are **open**. Tube feet do not bear suckers. Madreporite, spines and pedicellariae are absent. Some attach to the substratum by 'cirri'. Development includes **doliolaria larva**; it develops into a stalked, sessile **pentacrinoid larva** in feather stars.

Examples: *Ptilocrinus* (Sea lily with cirri), *Bathocrinus* (sea lily without cirri) and *Neometra* (feather star).

Sub-phylum-1: Eleutherozoa (Gr. *eleuthros* : free; *zoon* : animal)

They are free moving and do not have a stalk. Mouth (oral surface) is directed downwards. Madreporite is present. Ambulacral grooves are closed (except in *Asteroidea*).

The Sub-phylum **Eleutherozoa** is classified into **four classes 1. Asteroidea 2. Ophiuroidea 3. Echinoidea** and **4. Holothuroidea**.

Class-1: Asteroidea (Gr. *aster* : star; *eidos* : form)



Fig. 3.29 Starfish

This class includes sea stars (*star fishes*). The arms are not clearly demarcated from the central 'disc' of the body. **Ambulacral grooves** are open; tube feet bear suckers. Pedicellariae are two jawed and are useful in food capture, protection and cleaning the surface of the body. Madreporite is aboral (dorsal side) in position. Respiratory organs are **papulae** or **dermal branchiae**. Development includes **bipinnaria** and **brachiolaria larvae**.

Examples: *Asterias*, *Astropecten*, *Pentaceros*.

Class-2: Ophiuroidea (Gr. *ophiuros* : serpent tail; *eidos* : form)

This class includes *brittle stars*, *serpent stars*, *basket stars*, etc. Arms are long and clearly demarcated from the central disc of the body; arms are flexible, fragile and '**branched**' in some (*Gorgonocephalus*). They move by serpentine lashing of the arms.

Ambulacral grooves are closed; tube feet lack suckers. Madreporite is on the oral surface. Pedicellariae are absent. Anus is absent. Development includes **ophiopluteus larva**.

Examples: *Ophiothrix* (spiny brittle star), *Gorgonocephalus* (basket star).



Fig. 3.30 Brittle star

Class-3: Echinoidea (Gr. *echinos* : hedgehog; *eidos* : form)

This class includes 'sea urchins' such as, *heart urchins*, *sand dollars*, *sea biscuits*, *cake urchins* etc. The body is ovoid (*sea urchin*) or discoid (*sand dollars*) and covered with movable spines. Arms are absent;